

Notes on Conformal Field Theory

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Abstract

These are notes of conformal field theory. The primary objective of this note is to aid in the personal comprehension and formulation of the material presented in the book, rather than to generate profit or serve commercial purposes. It is not possible to present all the aspects of the theory. Given the limitations inherent in this process, occasional typographical errors may occur. The main referred textbook I take is [1]. Other references may be used but not mentioned. I use ChatGPT and DeepSeek to improve the text.

1 A Brief Introduction

As the foundation of the Standard Model, quantum field theory (QFT) has achieved remarkable success in describing and predicting physical phenomena. Numerous groundbreaking achievements, such as the prediction of the electron's anomalous magnetic moment [2], have confirmed its validity. However, despite its successes, QFT remains an incomplete theory and may not fully capture the fundamental principles of nature.

Conformal field theory (CFT) can be indeed viewed as “conformal-invariant quantum field theory”. It plays a central role in the description of phase transitions in two-dimensional systems, and in string theory, the attempt of unifying all forces of nature. Conformal transformations are nothing but dilations by a scaling factor that is a function of position. Two-dimensional conformal field theories are perfect examples of systems in which the symmetries are so powerful as to allow an exact solution of the problem. This feature, as well as the great variety of mathematical concepts needed in their solution and definition, have made conformal field theories one of the most active domains of research in mathematical physics.

There are quite a lot of motivations. Briefly speaking, we do not truncate energy scales in conformal field theory and it has profound mathematical structures. Here we simply give an example of critical phenomena described by Ising model.

Consider a lattice with atoms in interaction, such as in a solid. A simplified version of this interaction is embodied in the **Ising model**, in which the spins σ_i at the site i take only two values (+1 and -1) and the magnetic energy of the system is the sum over pairs of adjacent atoms:

$$E = \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \quad (1.1)$$

where the notation $\langle i, j \rangle$ indicates that summation is taken over pairs of nearest-neighbor lattice sites. The so-called **correlation length** ξ is defined as

$$\langle \sigma_i \sigma_j \rangle - \langle \sigma_i \rangle \langle \sigma_j \rangle \sim \exp \left(-\frac{i-j}{\xi(T)} \right). \quad (1.2)$$

Observable magnetic properties are directly derived from this.

At critical temperatures the correlation lengths diverge while zooming of the system does not obviously change the pattern. This independence of from microscopic details is **universality**. The behaviors of the system at critical points belong to a discrete set of **universality classes**, corresponding to different realizations of scale invariance. The critical points with the same critical parameters are described by the same conformal field theory.

In recent twenty years the main aspects where CFT dominates involve:

- String theory;
- Some gapless models in condensed matter theory;
- Entanglement entropy on Fermi surfaces;
- AdS/CFT correspondence;
- Topological order described by lower dimensional CFT.

Some of these will be discussed with concentration.

2 Statistical Mechanics

In this section we briefly introduce the basic knowledge on thermodynamics and statistical mechanics, which is profoundly related in many aspects of physics. Then, the emphasis is put on critical phenomena. In subsequent chapters, we will discuss renormalization group, a crucial and revolutionary concept.

If a system $\mathcal{S}$ is in thermal contact with its surroundings and hence is free to exchange energy with it, then all microstates of $\mathcal{S}$ do not have equal probabilities. This in turn provides us with a distribution of probabilities for microstates of $\mathcal{S}$. The probability that a specific microstate be the actual state of the system depends only on its energy, and is given by the **Boltzmann distribution**:

$$P_i = \frac{1}{Z} e^{-\beta E_i}, \quad \beta = \frac{1}{T}. \quad (2.1)$$

As known,

$$Z = \sum_i e^{-\beta E_i} \quad (2.2)$$

is the **partition function**, which serves as the normalization factor and plays crucial role as other thermodynamical quantities can be derived from it. The average energy of the system is given by

$$U = \frac{1}{Z} \sum_i E_i e^{-\beta E_i} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -T^2 \frac{\partial}{\partial T} \left(\frac{F}{T} \right), \quad (2.3)$$

where we have introduced the **free energy**:

$$F = -T \ln Z. \quad (2.4)$$

The **heat capacity** C at constant volume is

$$C = \left(\frac{\partial U}{\partial T} \right)_V = -T \frac{\partial^2 F}{\partial T^2}. \quad (2.5)$$

The limit $N \rightarrow \infty$ is called the **thermodynamic limit** since the variance of the macroscopic properties vanishes.

2.1 Classical Statistical Mechanics

In classical statistical mechanics, the best-known and simplest discrete model is Ising model, which we have introduced in previous chapter. Now we introduce the full version of it. On a lattice with N sites, each equipped with spin taking value -1 or 1 , the energy of a given configuration is

$$E = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i. \quad (2.6)$$

The first term in the energy represents the interaction of neighboring spins through a ferromagnetic ($J > 0$) or anti-ferromagnetic ($J < 0$). The second term represents the interaction with an external magnetic field h . In zero field ($h = 0$), the lowest energy configuration is doubly degenerate: The spins can be either all up ($+1$) or all down (-1). If the field h is nonzero, the lowest energy configuration will have all spins aligned with h .

A more realistic treatment is obtained by assuming the local spin to be a unit vector $\vec{n}$, with the configuration energy

$$E = J \sum_{\langle i,j \rangle} \vec{n}_i \cdot \vec{n}_j - \sum_i \vec{h} \cdot \vec{n}_i. \quad (2.7)$$

This is the **classical Heisenberg model**. It is often more convenient to replace the lattice by a continuum. The above Hamiltonian is then equivalent to

$$E = \int d^n x (J \partial_k \vec{n} \cdot \partial_k \vec{n} - \vec{h} \cdot \vec{n}). \quad (2.8)$$

The gradient term is equivalent to the nearest-neighbor interaction of the discrete case. For models defined on the continuum, the analogy between statistical mechanics and quantum field theory is manifest. The partition function of the ϕ^4 model is a sum over the possible configurations of the field ϕ :

$$\begin{aligned} Z &= \int \mathcal{D}\phi e^{-\beta E(\phi)} \\ &= \int \mathcal{D}\phi \exp \left(- \int d^n x \left(\frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} r \phi^2 + \frac{1}{4} u \phi^4 \right) \right). \end{aligned} \quad (2.9)$$

The partition function of a d -dimensional statistical model is thus entirely analogous to the generating functional of a quantum field in d spacetime dimensions in the Euclidean formalism.

2.2 Quantum Statistics

In quantum statistical mechanics, we must deal with quantum indeterminacy as well as the thermal fluctuations. We define the **density operator**

$$\rho = e^{-\beta H}, \quad (2.10)$$

where H is the Hamiltonian of the system. The partition function may be expressed as a sum over eigenstates of H :

$$Z = \sum_n e^{-\beta E_n} = \text{Tr}(\rho). \quad (2.11)$$

The statistical average of an operator A is then

$$\langle A \rangle = \sum_n \langle n | e^{-\beta E_n} A | n \rangle = \text{Tr}(\rho A). \quad (2.12)$$

2.3 Critical Phenomena

Phase transitions are probably the most interesting feature of statistical systems. They are often characterized by a sudden change in the macroscopic properties. We distinguish **first order transitions** from continuous transitions. First order transitions are characterized by a jump in the energy U at critical temperature. Sometimes we refer to this as **latent heat**. This means the temperature transiently stops changing when system delivers or absorbs a certain amount of energy. Continuous phase transitions does not involve any latent heat. However, the derivative of such quantities may be discontinuous.

We are now more interested in continuous phase transitions because of their relation to conformal invariance. The two-dimensional Ising model exhibits such a transition. The critical temperature T_c of Ising model is given by

$$\sinh \left(\frac{2J}{T_c} \right) = 1. \quad (2.13)$$

Above critical temperature, the spontaneous magnetization ($h = 0$) vanishes, while below T_c the magnetization takes a nonzero value. As temperature rises, M approaches zero according to the

power law

$$M \sim (T_c - T)^{1/8}. \quad (2.14)$$

The system is then in its **ferromagnetic phase**. Obviously M is continuous in such a transition, whereas the derivative of it diverges:

$$\chi = \frac{\partial M}{\partial h} \sim (T_c - T)^{-7/4}. \quad (2.15)$$

Meanwhile, the correlation length diverges as

$$\xi(T) \sim \frac{1}{T - T_c}, \quad (2.16)$$

which agrees with our intuition that the transition occurs over the scale of whole system.

2.4 Broken Symmetry

By broken symmetry, we mean that a symmetry of the configuration energy that is no longer reflected in the macrostate of the statistical system. For instance, as stated, the magnetization $\langle \sigma_i \rangle$ is nonzero in the low temperature phase of the Ising model in the zero field, and the spin reversal symmetry is then broken. Here we call the magnetization the **order parameter**, as its value exactly describes the phase transition, and it's the simplest.

3 Quantum Field Theory

We have emphasized the significance of quantum field theory as the fundamental framework of modern physics, particularly in the realm of high-energy physics. Therefore, it is essential to briefly review its key aspects. In this section, we will first introduce two fundamental formalisms: canonical quantization and the path integral approach. We will then examine the core elements that govern our theory, including the correlation function, symmetry transformations, the energy-momentum tensor, and other relevant concepts.

3.1 Quantum Fields

3.2 Path integral

The path integral formalism is fully equivalent to the canonical quantization approach we have previously discussed. In fact, quantum field theory can be entirely formulated within the path integral framework if desired. While the path integral formalism presents certain challenges, particularly in the computation of scattering amplitudes, it is often more advantageous than canonical quantization when it comes to evaluating correlation functions.

We should already be familiar with the path integral formulation in quantum mechanics. In the well-known double-slit experiment, the total amplitude for the two electrons is given by the sum of the contributions from each path:

$$A = A_1 + A_2. \quad (3.1)$$

Imagine a scene where infinitely many slits are distributed on infinitely many boards, the total amplitude would be calculated the same as before.

In quantum mechanics, the Hamiltonian is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}, t). \quad (3.2)$$

Start with the initial $|i\rangle = |x_i\rangle$ and final state $|f\rangle = |x_f\rangle$, if the Hamiltonian is time-independent, the amplitude would be

$$\langle f|i\rangle = \langle x_f|e^{-i\hat{H}t}|x_i\rangle. \quad (3.3)$$

Otherwise, i.e. the Hamiltonian is time-dependent, we are in a complicated situation. We make time slices $t_j = t_i + \delta t$, and insert a complete set of position eigenstates $\int dx_n |x_n\rangle \langle x_n|$ for each slice, then the amplitude is

$$\langle f|i\rangle = \int dx_n \cdots dx_1 \langle x_f|e^{-i\hat{H}(t)\delta t}|x_n\rangle \langle x_n|e^{-i\hat{H}(t)\delta t}|x_{n-1}\rangle \cdots \langle x_2|e^{-i\hat{H}(t)\delta t}|x_1\rangle \langle x_1|e^{-i\hat{H}(t)\delta t}|x_i\rangle. \quad (3.4)$$

Consider a specific slice:

$$\begin{aligned} \langle x_{j+1}|e^{-i\hat{H}(t)\delta t}|x_j\rangle &= \int \frac{dp}{2\pi} \langle x_{j+1}|p\rangle \langle p|\exp\left(-i\left(\frac{\hat{p}^2}{2m} + V(x_j, t_j)\right)\delta t\right)|x_j\rangle \\ &= \exp(-iV(x_j, t_j)\delta t) \int \frac{dp}{2\pi} \exp\left(-i\frac{\hat{p}^2}{2m}\delta t\right) \exp(i\hat{p}(x_{j+1} - x_j)) \\ &= N \exp(-iV(x_j, t_j)\delta t) \exp\left(\frac{im\delta t}{2} \frac{(x_{j+1} - x_j)^2}{(\delta t)^2}\right) \\ &= N \exp(i\mathcal{L}(x, \dot{x})). \end{aligned} \quad (3.5)$$

We have used $\langle p|x\rangle = e^{ipx}$ and an important Gaussian integral

$$\int dp e^{-ap^2+Jp} = \sqrt{\frac{2\pi}{a}} e^{J^2/2a}$$

in the equation above. Each path contribute to the same amplitude, but a different phase that is proportional to the action S .

Finally, we end up with the total amplitude

$$\langle f|i\rangle = N^n \int dx_n \cdots dx_1 \exp(i\mathcal{L}(x_n, \dot{x}_n)) \cdots \exp(i\mathcal{L}(x_1, \dot{x}_1)). \quad (3.6)$$

Under the limit $\delta t \rightarrow 0$, we get

$$\langle f|i\rangle = N^n \int_{x_i}^{x_f} \mathcal{D}x(t) e^{iS[x(t)]}. \quad (3.7)$$

In quantum field theory, we are dealing with path integral from vacuum states.

$$\langle 0|0\rangle = \int \mathcal{D}\Phi_n(x) \cdots \mathcal{D}\Phi_1(x) \langle 0|e^{-i\delta t \hat{H}(t_n)}|\Phi_n\rangle \langle \Phi_n| \cdots \langle \Phi_1|e^{-i\delta t \hat{H}(t_1)}|0\rangle, \quad (3.8)$$

$$\begin{aligned} \langle \Phi_{j+1}|e^{-i\hat{H}(t_n)}|\Phi_j\rangle &= \int \mathcal{D}\Pi_j \langle \Phi_{j+1}|\Pi_j\rangle \langle \Pi_j|\exp\left(-i\delta t \int d^3x \left(\frac{\hat{\pi}^2}{2} + V(\hat{\Phi})\right)\right)|\Phi_j\rangle \\ &= N \exp\left(i\delta t \int d^3x \mathcal{L}[\phi_i, \partial_t \phi_j]\right). \end{aligned} \quad (3.9)$$

In equations above, the symbol $\mathcal{D}\Phi$ we introduced means the **Lebesgue measure** on the space of paths, it is difficult to define it in a mathematically rigorous way. Here one must use ingenuity to work with these formal expressions. [3][4] are recommended textbooks on quantum field theory.

3.3 Correlation Functions

3.4 Wick's Theorem

We have introduce two special orderings on the field operators: normal ordering, which places all annihilation operators on the right, and time ordering, which sorts operators in chronological order. **Wick's theorem** links these two orderings together.

We must first define the **contraction** of two operators within a normal order. The contraction of two operators ϕ_i and ϕ_j is simply to "package" them into a Feynman propagator, or say, a correlation function $\langle \phi_i \phi_j \rangle$. The theorem itself states: The time-ordered product is equal to the normal-ordered product, plus all possible ways of contracting pairs of fields within it. For instance,

$$\begin{aligned} \mathcal{T}(\phi_1 \phi_2 \phi_3 \phi_4) &= : \phi_1 \phi_2 \phi_3 \phi_4 : + : \overbrace{\phi_1 \phi_2} \phi_3 \phi_4 : + : \overbrace{\phi_1 \phi_3} \phi_2 \phi_4 : + : \overbrace{\phi_1 \phi_4} \phi_2 \phi_3 : + \\ &+ : \overbrace{\phi_2 \phi_3} \phi_1 \phi_4 : + : \overbrace{\phi_2 \phi_4} \phi_1 \phi_3 : + : \overbrace{\phi_3 \phi_4} \phi_1 \phi_2 : + : \overbrace{\phi_1 \phi_2} \overbrace{\phi_3 \phi_4} : + \\ &+ : \overbrace{\phi_1 \phi_3} \overbrace{\phi_2 \phi_4} : + : \overbrace{\phi_1 \phi_4} \overbrace{\phi_2 \phi_3} : + : \overbrace{\phi_2 \phi_3} \overbrace{\phi_1 \phi_4} : + : \overbrace{\phi_2 \phi_4} \overbrace{\phi_1 \phi_3} : + \\ &+ : \overbrace{\phi_3 \phi_4} \overbrace{\phi_1 \phi_2} : + : \overbrace{\phi_1 \phi_2} \overbrace{\phi_3 \phi_4} : . \end{aligned} \quad (3.10)$$

Wick's theorem also applies to free fermions, with the difference that a minus sign must be included in front of each term, according to the number of anti-commutations required to bring the contracted fields next to each other. For instance, application of equation (3.10) to the fermion fields

becomes

$$\begin{aligned}
\mathcal{T}(\psi_1\psi_2\psi_3\psi_4) = & : \psi_1\psi_2\psi_3\psi_4 : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + \\
& : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + \\
& : \overbrace{\psi_1\psi_2\psi_3\psi_4} : + : \overbrace{\psi_1\psi_2\psi_3\psi_4} : .
\end{aligned} \tag{3.11}$$

The proof of Wick's theorem will not be given here.

4 Conformal Invariance

4.1 Conformal Transformation

Do you remember studying elementary geometry, or topics like “properties of shapes,” during your primary and middle school years? To introduce the concept of conformal transformations and conformal symmetry, let us start from this familiar ground.

For simplicity, let us consider two-dimensional plane figures. What does it mean for two figures to be congruent? A primary school textbook defines congruence as follows: “Figures that overlap perfectly (having the same shape and size) are congruent.” The phrase “overlap perfectly” suggests that some operations are applied to one figure to align it with the other. What operations are we considering here? The same textbook states that these operations include “shifting, rotating, and flipping.” In mathematical terms, this means that congruence is defined by transformations involving translations, rotations, and reflections.

Why do we adopt the definition of “considering objects as the same if they can be transformed into one another by a given set of operations”? Since this definition is understandable even for primary school students, it must be based on our everyday intuition. Indeed, human perception remains invariant under translations, rotations, and reflections. Consequently, it feels natural to regard objects related by these transformations as identical. But why is our perception invariant under these transformations? Although delving deeper into this question enters the realm of evolutionary biology, a plausible hypothesis is that since the physical world itself is inherently invariant under translations, rotations, and reflections, possessing such a perception has conferred an evolutionary advantage.

Mathematically speaking, the set of symmetries in Euclidean geometry, including translations and rotations, forms a group known as the Euclidean group. This group is a Lie group generated by infinitesimal transformations, and it is well-known in physics that the generators correspond to linear momentum and angular momentum. Since translations and rotations together form a closed group, it is optional whether to include reflections, which stand as isolated elements in this structure. In other words, both the set {translations, rotations} and the set {translations, rotations, reflections} are independently closed under composition. However, our physical world does not exhibit reflection symmetry at a fundamental level, for instance, due to weak interactions in particle physics. In fact, when writing this, I assumed that “congruence” does not include reflections, but upon reviewing an elementary school textbook, I was surprised to see that it does.

Similarity Transformations and Dilations

Next, let us consider similarity transformations. A middle school textbook defines similarity as follows: “Two figures are similar if one is congruent to a scaled version of the other with a fixed ratio.” Here, we encounter the first instance of a conformal transformation: the dilation (scaling transformation). If we include the case where the scaling factor is 1, we see that congruent figures are necessarily similar. Moreover, note that the definition of similarity is built upon the definition of congruence. Mathematically, translations, rotations, reflections, and dilations together form a closed group. However, just as before, the inclusion of reflections does not affect the group’s closure. Given that our physical world is not reflection-invariant, as previously mentioned, we define the “scaling transformation group” as the group generated by translations, rotations, and dilations (excluding reflections).

The principle behind similarity is that objects related by similarity transformations should be considered equivalent. The naturalness of this definition aligns with our everyday intuition, which has been shaped by evolution. For instance, if we were unable to recognize the same letter at different sizes or rotations, it would be highly inconvenient. More critically, in a survival scenario, failing to recognize a lion at different distances or orientations would be disastrous—such a trait would have

led to extinction long before humans developed literacy.

Inversion Transformations

Having covered elementary and middle school topics, let us now introduce a concept sometimes taught in high school: inversion. An inversion transformation maps lines and circles to other lines and circles. While the center of inversion can be arbitrary, consider the specific case of an inversion centered at the origin. Mathematically, the inversion of a position vector $\vec{x}$ is defined as

$$\vec{x} \rightarrow \frac{\vec{x}}{|\vec{x}|^2}. \quad (4.1)$$

Under this transformation, a circle given by $(\vec{x} - \vec{a})^2 = r^2$ is mapped to another circle described by

$$(\vec{a}^2 - r^2)|\vec{x}|^2 - 2\vec{a} \cdot \vec{x} + 1 = 0. \quad (4.2)$$

For example, the unit circle $|\vec{x}| = 1$ remains invariant under inversion centered at the origin.

A classic university entrance exam problem related to inversion is as follows: In the complex plane, define $w = \frac{1}{z}$ for all points z except the origin. Given a nonzero complex number α , let L be the perpendicular bisector of the line segment connecting α and the origin. As z moves along L , the trajectory of w forms a circle with one point removed, see Figure 4.1. Find the center and radius of this circle.

Although a detailed solution is omitted here, note that in the complex plane, the inversion transformation defined in (4.1) is given by $w = 1/\bar{z}$. Hence, the transformation in this problem is a combination of inversion and complex conjugation, which corresponds to reflection with respect to the real axis.

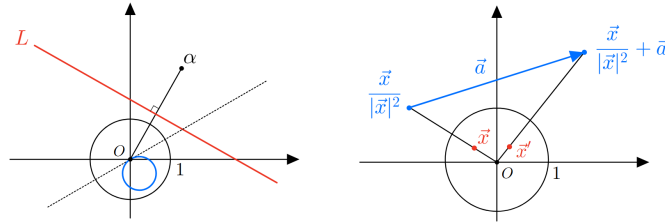


Figure 1: (Left) A problem about inversion. (Right) Special conformal transformation.

Special Conformal Transformations

So far, we have introduced transformations—translations, rotations, and dilations that preserve lines and circles. Are these transformations closed under composition? To investigate this, consider the operation of an inversion, followed by a translation, followed by another inversion, see Figure 1:

$$\vec{x} \rightarrow \frac{\vec{x}}{|\vec{x}|^2} \rightarrow \frac{\vec{x}}{|\vec{x}|^2} + \vec{a} \rightarrow \frac{\frac{\vec{x}}{|\vec{x}|^2} + \vec{a}}{\left| \frac{\vec{x}}{|\vec{x}|^2} + \vec{a} \right|^2}. \quad (4.3)$$

Simplifying the right-hand side, we obtain

$$\vec{x} \rightarrow \frac{\vec{x} + \vec{a}|\vec{x}|^2}{1 + 2\vec{a} \cdot \vec{x} + |\vec{a}|^2|\vec{x}|^2}. \quad (4.4)$$

This transformation is in stark contrast to translations, rotations, and dilations, all of which are linear transformations of coordinates. Unlike these transformations, special conformal transformations

cannot be constructed from their combinations alone. The special conformal transformation, which is the focus of this lecture, depends on a parameter vector $\vec{a}$.

From its construction, it is evident that special conformal transformations map straight lines and circles to straight lines and circles. Equivalently, they preserve the angles between two intersecting lines (or arcs of circles). Thus, a special conformal transformation can be understood as a local dilation transformation at each point, maintaining angle preservation.

Transformations that preserve angles are called conformal mappings. While general conformal mappings do not always form a group, an important observation is that the set of transformations consisting of translations, rotations, dilations, and special conformal transformations forms a closed group, known as the **conformal group**, and the transformations it generates are called **conformal transformations**. It is worth noting that the group remains closed even without including the inversion operation used to introduce special conformal transformations. Therefore, in this lecture, we do not include inversion as part of the conformal transformation group. Indeed, in the real world, the conformal symmetry that manifests does not include inversion operations.

Compared to translations, rotations, and dilations, which appear in congruence and similarity transformations, inversions and special conformal transformations are not as intuitive. If one were given a test to recognize equivalent shapes under special conformal transformations, I would not be confident in achieving a high accuracy. Unlike Euclidean transformations and scale transformations, our perception does not appear to be invariant under conformal transformations. Nevertheless, the reason we study conformal symmetry in this lecture is that remarkable phenomena in nature exhibit this symmetry. In fact, a fundamental observation is that many physical systems exhibit conformal symmetry in extreme conditions. We will explore this concept further in the following sections.

Now we discuss conformal transformations. A scaling transformation

$$g'_{\mu\nu}(x) = \Lambda(x)g_{\mu\nu}(x) \quad (4.5)$$

is referred to as a **conformal transformation**, $\Lambda(x)$ is the **conformal factor**. Under a infinitesimal transformation $x_\mu \rightarrow x_\mu + \epsilon_\mu$, the metric transforms as

$$g_{\mu\nu} \rightarrow g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta} = (\delta_\mu^\alpha - \partial_\mu \epsilon^\alpha)(\delta_\nu^\beta - \partial_\nu \epsilon^\beta) g_{\alpha\beta} = g_{\mu\nu} - \partial_\mu \epsilon_\nu - \partial_\nu \epsilon_\mu. \quad (4.6)$$

Therefore,

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = f(x)g_{\mu\nu}, \quad (4.7)$$

where

$$f(x) = \Lambda(x) - 1 = \frac{2}{d} \partial_\rho \epsilon^\rho. \quad (4.8)$$

We primarily concentrate on conformal field theory in 2-dimensional space, then

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = f(x)\eta_{\mu\nu}. \quad (4.9)$$

Applying ∂_ρ on the both sides and making use of displacement symmetry, we obtain

$$2\partial_\rho \partial_\mu \partial_\nu = (\eta_{\mu\rho} \partial_\nu + \eta_{\nu\rho} \partial_\mu - \eta_{\mu\nu} \partial_\rho) f \quad (4.10)$$

Contracting it using $\eta^{\mu\nu}$, then

$$2\partial^2 \epsilon_\mu = (2 - d)\partial_\mu f. \quad (4.11)$$

Furthermore, act ∂_ρ on the equation above, it is straightforward to see that

$$(d - 1)\partial^2 f = 0. \quad (4.12)$$

This agrees with intuition that any transformation on 1-dimensional space is conformal; For 2-dimensional case, there are infinitely many solutions, which is why it is of significance. With the above two equations, we have

$$\partial^2 \partial \cdot \epsilon = 0. \quad (4.13)$$

Thus

$$\epsilon_\mu = a_\mu + b_{\mu\nu} x^\nu + c_{\mu\nu\rho} x^\nu x^\rho, \quad c_{\mu\nu\rho} = c_{\mu\rho\nu}. \quad (4.14)$$

The three terms in the r.h.s of the equation above account for transformation, dilation and special conformal transformation, respectively.

4.2 Conformal Groups and Algebras

Conformal transformations form a group. For now we examine generators of conformal transformations in d -dimensional Euclidean space. For example, an infinitesimal translation, transforms a function $f(x^\mu)$ as follows under an infinitesimal constant vector ϵ^μ :

$$f(x^\mu) \rightarrow f(x^\mu + \epsilon^\mu) = f(x^\mu) + \epsilon^\mu \partial_\mu f(x^\mu). \quad (4.15)$$

The equation above is exactly the result of Taylor expansion. Thus, the generator of translations is given by

$$\mathcal{P}_\mu = -i\partial_\mu. \quad (4.16)$$

Similarly, the generator of rotations is obtained by parameterizing infinitesimal rotations with an infinitesimal constant antisymmetric tensor $\epsilon^{\mu\nu} = -\epsilon^{\nu\mu}$. The function transforms as:

$$\begin{aligned} f(x^\mu) &\rightarrow f(x^\mu - \epsilon^{\mu\nu} x_\nu) = f(x^\mu) - \epsilon^{\mu\nu} x_\nu \partial_\mu f(x^\mu) \\ &= f(x^\mu) + \frac{1}{2} \epsilon^{\mu\nu} (x_\mu \partial_\nu - x_\nu \partial_\mu) f(x^\mu), \end{aligned} \quad (4.17)$$

giving the generator of rotations:

$$\mathcal{J}_{\mu\nu} = -i(x_\mu \partial_\nu - x_\nu \partial_\mu). \quad (4.18)$$

Furthermore, the generator of dilations is derived as follows:

$$f(x^\mu) \rightarrow f(x^\mu - \epsilon x^\mu) = f(x^\mu) - \epsilon x^\mu \partial_\mu f(x^\mu), \quad (4.19)$$

leading to the generator

$$D_\mu = -ix^\mu \partial_\mu. \quad (4.20)$$

Finally, for an infinitesimal constant vector a^μ , ignoring second-order infinitesimals, we obtain:

$$x^\mu \rightarrow x^\mu + a^\mu x^2 - 2(a^\rho x_\rho) x^\mu, \quad (4.21)$$

Therefore, the generator of special conformal transformations is given by

$$\mathcal{K}_\mu = i(2x_\mu x^\nu \partial_\nu - x^2 \partial_\mu). \quad (4.22)$$

Conformal transformations form a simple Lie algebra. To verify this, let us compute the commu-

tation relations of the generators. First, for the Euclidean subgroup, we have:

$$\begin{aligned}
i[\mathcal{J}_{\mu\nu}, \mathcal{J}_{\rho\sigma}] &= \delta_{\nu\rho}\mathcal{J}_{\mu\sigma} - \delta_{\mu\rho}\mathcal{J}_{\nu\sigma} - \delta_{\sigma\mu}\mathcal{J}_{\rho\nu} + \delta_{\sigma\nu}\mathcal{J}_{\rho\mu}, \\
i[\mathcal{P}_\mu, \mathcal{J}_{\rho\sigma}] &= \delta_{\mu\rho}\mathcal{P}_\sigma - \delta_{\mu\sigma}\mathcal{P}_\rho, \\
i[\mathcal{P}_\mu, \mathcal{P}_\nu] &= 0.
\end{aligned} \tag{4.23}$$

Similarly, for the relations involving dilations and special conformal transformations:

$$\begin{aligned}
i[\mathcal{P}_\mu, \mathcal{D}] &= -\mathcal{P}_\mu, \\
i[\mathcal{J}_{\mu\nu}, \mathcal{D}] &= 0, \\
i[\mathcal{K}_\mu, \mathcal{D}] &= \mathcal{K}_\mu, \\
i[\mathcal{P}_\mu, \mathcal{K}_\nu] &= 2\delta_{\mu\nu}\mathcal{D} + 2\mathcal{J}_{\mu\nu}, \\
i[\mathcal{K}_\mu, \mathcal{K}_\nu] &= 0, \\
i[\mathcal{K}_\mu, \mathcal{J}_{\rho\sigma}] &= \delta_{\mu\rho}\mathcal{K}_\sigma - \delta_{\mu\sigma}\mathcal{K}_\rho.
\end{aligned} \tag{4.24}$$

It can be easily seen that the conformal algebra in Euclidean space is closed.

The conformal algebra in Euclidean space can be identified with the symmetry algebra of $(d+1)$ -dimensional Minkowski space, specifically $SO(d+1, 1)$. To see this, we define the generators of $SO(d+1, 1)$ as follows:

$$\begin{aligned}
M_{d+1, \mu} &= \frac{-\mathcal{P}_\mu + \mathcal{K}_\mu}{2}, \\
M_{d+2, \mu} &= \frac{-\mathcal{P}_\mu - \mathcal{K}_\mu}{2}, \\
M_{\mu\nu} &= \mathcal{J}_{\mu\nu}, \\
M_{d+2, d+1} &= \mathcal{D}.
\end{aligned} \tag{4.25}$$

These generators satisfy the commutation relations of $SO(d+1, 1)$, given by:

$$i[M_{AB}, M_{CD}] = \eta_{BC}M_{AD} - \eta_{AC}M_{BD} - \eta_{AD}M_{CB} + \eta_{BD}M_{CA}. \tag{4.26}$$

Thus, the conformal algebra of d -dimensional Euclidean space is isomorphic to $SO(d+1, 1)$, the Lorentz algebra in $\mathbb{R}^{d+1, 1}$ Minkowski spacetime, confirming that it possesses

$$\binom{d+2}{2} = \frac{(d+2)(d+1)}{2}$$

generators.

From the algebraic perspective, the difference between $SO(d+1, 1)$ and $SO(d+2)$ is minimal. By replacing $x^{d+2} \rightarrow ix^0$, the metric in the commutation relations changes from δ_{AB} to η_{AB} , leading to the transition from $SO(d+2)$ to $SO(d+1, 1)$. Since the unitary representations of these algebras are significant, especially in physics, the reality properties of the generators are crucial. The unitary representations of these algebras differ fundamentally; for instance, $SO(d+2)$ possesses unitary representations of all dimensions, while $SO(d+1, 1)$ and $SO(d, 2)$ lack certain well-defined finite-dimensional unitary representations.

5 Basics in Conformal Field Theory

5.1 Conformal Ward-Takahashi Identity

For an arbitrary infinitesimal conformal transformation $x_\mu \rightarrow x_\mu + \epsilon_\mu$, we have

$$\partial_\mu(\epsilon_\nu T^{\mu\nu}) = \epsilon_\mu(\partial_\mu T^{\mu\nu}) + \frac{1}{2}(\partial_\rho \epsilon^\rho) \eta_{\mu\nu} T^{\mu\nu} + \frac{1}{2}\epsilon^{\alpha\beta} \partial_\alpha \epsilon_\beta T^{\mu\nu}. \quad (5.1)$$

Quite similar to equation (4.14), the three terms on the r.h.s of the equation above account for transformation, dilation and special conformal transformation, respectively. The second term corresponds to $f(x)$ in equation (4.7).

5.2 Primary Fields and Descendants

We have to determine how fields transform under conformal transformations. We assume that physical states are eigenstates of l_0 and $\bar{l}_0$ with eigenvalues $\{h, \bar{h}\}$, known as the **conformal dimensions**. Under a scaling transformation $z \rightarrow \lambda z$, $\bar{z} \rightarrow \bar{\lambda} \bar{z}$, the field operators transform as

$$\phi'(\lambda z, \bar{\lambda} \bar{z}) \rightarrow \lambda^{-h} \bar{\lambda}^{-\bar{h}} \phi(z, \bar{z}). \quad (5.2)$$

If a field transforms under an arbitrary conformal map $SL(2, \mathbb{C})$: $z \rightarrow w(z)$, $\bar{z} \rightarrow \bar{w}(\bar{z})$,

$$\phi'(w, \bar{w}) = \left(\frac{dw}{dz}\right)^{-h} \left(\frac{d\bar{w}}{d\bar{z}}\right)^{-\bar{h}} \phi(z, \bar{z}), \quad (5.3)$$

then it is called a **quasi-primary field**. Under infinitesimal transformation $z \rightarrow z' = z + \epsilon z$, we obtain the infinitesimal transformation of quasi-primary field:

$$\delta\phi = \phi'(z', \bar{z}') - \phi(z, \bar{z}) = -(h\phi\partial\epsilon + \epsilon\partial\phi) - (\bar{h}\phi\bar{\partial}\bar{\epsilon}) + (\bar{\epsilon}\bar{\partial}\phi). \quad (5.4)$$

A field whose transformation is given by this equation is called a **primary field**. The key distinction between these two concepts is that a primary field satisfy conformal invariance under *local* conformal transformations; However, a quasi-primary field is weakly-constraint that it transforms under *global* conformal transformations, which is generated solely by special linear group $SL(2, \mathbb{C})$. If a field is not primary, it is generally called **secondary**.

The definition (5.3) imposes strict demand on the form of correlation functions, which is exactly why 2-dimensional conformal field theory possess excellent properties. We will discuss this immediately in the next section.

5.3 Correlation Functions in 2-Dimensional CFT

Generally, a field theory is defined by a Lagrangian. Furthermore, we get used to obtain correlation functions by doing functional derivative on the generating functional. In other words, path integral formalism is all we need. However, the symmetry of 2-dimensional conformal field theory imposes strong constraints on correlation functions, so that correlation functions can be written down directly from symmetry requirements. We shall discuss with concentration for two-point function, and give other results in a straightforward way.

First, consider the two-point function

$$g_{ij}(x_1, x_2) = \langle 0 | O_i(x_1) O_j(x_2) | 0 \rangle. \quad (5.5)$$

By translation invariance, we have

$$0 = (\partial_\mu^1 + \partial_\mu^2) g_{ij}(x_1, x_2). \quad (5.6)$$

This implies the function depends solely on the difference of coordinates $x_1 - x_2$. By rotational invariance,

$$\begin{aligned} 0 &= ((x_\mu^1 \partial_\nu^1 - x_\nu^1 \partial_\mu^1) + (x_\mu^2 \partial_\nu^2 - x_\nu^2 \partial_\mu^2)) g_{ij}(x_1 - x_2) \\ &= ((x_\mu^1 - x_\mu^2) \partial_\nu^1 - (x_\nu^1 - x_\nu^2) \partial_\mu^1) g_{ij}(x_1 - x_2). \end{aligned} \quad (5.7)$$

This implies the function depends on the distance between the points, which means $g_{ij}(x_1 - x_2) = g_{ij}(|x_1 - x_2|)$. Next, the conservation of dilations gives

$$0 = (x_\mu^1 \partial_\mu^1 + x_\mu^2 \partial_\mu^2 + \Delta_i + \Delta_j) g_{ij}(|x_1 - x_2|). \quad (5.8)$$

This means that the function is homogeneous of degree $-(\Delta_i + \Delta_j)$:

$$g_{ij}(x_1, x_2) = \frac{c_{ij}}{|x_1 - x_2|^{\Delta_i + \Delta_j}}. \quad (5.9)$$

Finally, we consider the invariance of $\mathcal{K}_\mu$. Applying (4.22) on the correlation function, we arrive at:

$$0 = (\Delta_1 - \Delta_2)(x_\mu^1 - x_\mu^2)|x_1 - x_2|^{-\Delta_1 - \Delta_2 - 2}. \quad (5.10)$$

The equation holds if and only if $\Delta_1 = \Delta_2$. In other words, the two-point correlation function vanishes unless fields possess the same conformal dimension. Note that this conclusion does not dominate descendant operators. For instance, correlation function $(\langle O_1(x_1) \partial^\mu O_2(x_2) \rangle)$ vanishes when $\Delta_1 = \Delta_2 = (d-2)/2$. But generally, it does not.

In a similar manner, due to the conformal invariance, the form of three-point correlation functions are determined, up to a constant c_{ijk} :

$$c_{ijk}(x_1, x_2, x_3) = \frac{c_{ijk}}{x_{12}^{\Delta_i + \Delta_j - \Delta_k} x_{23}^{\Delta_j + \Delta_k - \Delta_i} x_{31}^{\Delta_k + \Delta_i - \Delta_j}}. \quad (5.11)$$

Polyakov was among the first to point out that behind (2-dimensional) critical phenomena lies not only scale invariance but also conformal symmetry. The first check of his argument was the calculation of the 3-point function for the exact solution of the 2-dimensional critical Ising model, known as Onsager's solution, which confirmed that it takes the form predicted by conformal field theory[5].

As our final example of correlation functions, we are faced with four-point function. However, things may be different from the three-point case that the function is not determined, due to the **cross ratio**:

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \quad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}. \quad (5.12)$$

They are dimensionless. Moreover, they are invariant under all conformal transformations. Therefore, any correlation function multiplied by a cross ratio will also be conformal-invariant. Often we write the four-point function as the form:

$$\langle O_1(x_1) O_2(x_2) O_3(x_3) O_4(x_4) \rangle = \frac{g(u, v)}{x_{12}^{\Delta_1 + \Delta_2} x_{34}^{\Delta_3 + \Delta_4}} \left(\frac{x_{14}}{x_{23}} \right)^{\Delta_1 + \Delta_2} \left(\frac{x_{24}}{x_{13}} \right)^{\Delta_3 + \Delta_4}. \quad (5.13)$$

There is no further constraints on function $g(u, v)$. Imposing the conditions from operator product expansion is one of the core courses of modern conformal field theory, with assistance from modern

techniques, such as the quantum Monte-Carlo simulation. This procedure is known as the **conformal bootstrap**. This will be included in subsequent chapters.

5.4 Operator Product Expansion

In conformal field theory, the terminology of “field” might be slightly different from what we have met in quantum field theory. In QFT, we refer to objects ϕ that lie in the action and to be integrated over in the path integral as field. In CFT, the term “field” refers to any local expression that we can write down, such as $\partial\phi$ or $e^{i\phi}$. However, this does not mean that our theories are becoming more complicated, so there’s nothing to be afraid of.

We now define the **operator product expansion (OPE)**, one of the most important techniques in conformal field theory. It is a statement about what happens as local operators approach each other. The idea is that two local operators inserted at nearby points can be closely approximated by a string of operators at one of these points. Let’s denote all the local operators of the CFT by $\mathcal{O}_i$, where i runs over the set of all operators. Then the OPE is

$$\mathcal{O}_i(z, \bar{z})\mathcal{O}_j(w, \bar{w}) = \sum_k C_{ij}^k(z-w, \bar{z}-\bar{w})\mathcal{O}_k(w, \bar{w}), \quad (5.14)$$

where $C_{ij}^k(z-w, \bar{z}-\bar{w})$ depends only on the separation between the two operators, like the constraint due to translation invariance on the two-point correlation function. The significance of defining such expansions is that now it is possible to express a string of operators in a simpler way.

When other operators are inserted, the following relation holds

$$\langle \mathcal{O}_i(z, \bar{z})\mathcal{O}_j(w, \bar{w}) \cdots \rangle = \sum_k C_{ij}^k(z-w, \bar{z}-\bar{w}) \langle \mathcal{O}_k(w, \bar{w}) \cdots \rangle, \quad (5.15)$$

Note that the operators are always assumed to be time-ordered, which means that operators commute in the correlation function. If the operators are Grassmann variables, they pick up a minus sign once commuted.

We have seen in the previous section, that conformal symmetry determined some correlation functions. A theory is considered **explicitly solvable** if all of its correlators can be solved. This can indeed be achieved using operator algebra (5.14), as it allows for a recursive application in a nested manner.

5.5 Radial Quantization and Virasoro Algebra

Hilbert space and operators are nonetheless extremely useful in conformal field theory because of the power of algebraic and group-theoretical methods. We always distinguish a time direction from space direction. This is natural in Minkowski space, in which we live. Nevertheless, this is somewhat arbitrary for Euclidean space. The choice of space and time leads to the so-called **radial quantization** in two-dimensional conformal field theory.

We may initially define our theory on a cylinder, with time t going from $-\infty$ to $+\infty$ along the flat direction of the cylinder, and space being compactified with a coordinate x going from 0 to L . The cylinder can be described by the complex coordinate $\xi = t + ix$, or $\xi = t - ix$ equivalently. Then we define the map

$$z = e^{2\pi\xi/L}. \quad (5.16)$$

Now, the remote past ($t \rightarrow -\infty$) is situated at $z = 0$ while the remote future ($t \rightarrow +\infty$) is at $z = \infty$. We also assume the existence of vacuum state $|0\rangle$ upon which a Hilbert space is constructed by

application of creation operators. In free theories, the vacuum may be defined as the state annihilated by the positive frequency part of the field.

The quantum generators L_n obey a similar algebra:

$$[L_n, L_m] = (n - m)L_{n+m} + \frac{c}{12}n(n^2 - 1)\delta_{n+m,0}, \quad (5.17)$$

where c is the central charge of the theory. This is the celebrated **Virasoro algebra**.

A mathematician will refer to the presence of c in Virasoro algebra (5.17) as **central extension**. Indeed, c leads to **conformal anomaly** and it describes the way a system reacts to macroscopic length scales introduced. If a CFT is defined on the boundary of a cylinder, it gives rise to Casimir energy, the change brought about by the periodicity condition on the cylinder; If it is defined on curved two-dimensional manifold, the curvature introduces a nonzero expectation value of energy-momentum tensor, which is proportional to curvature R and central charge c . This breaking of scale invariance is known as **trace anomaly**. In string theory, the neutralization of trace anomalies directly gives the dimensions of bosonic and supersymmetric string, 26 and 10, respectively.

Radial Ordering

Radial ordering is usually used for Euclidean space. Combined with the discussion on radial quantization, the radial ordering can be defined as

$$\mathcal{R}\phi_1(z)\phi_2(w) = \begin{cases} \phi_1(z)\phi_2(w), & |z| > |w| \\ \phi_2(w)\phi_1(z), & |z| < |w| \end{cases}, \quad (5.18)$$

To be concise the symbol $\mathcal{R}$ is omitted in the texts.

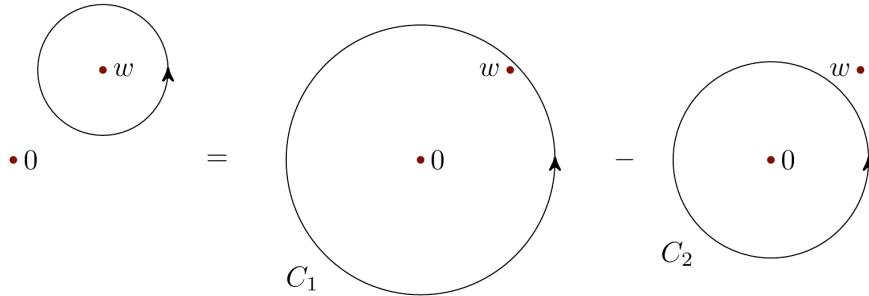


Figure 2: Radial ordering of operators.

With the help of contour integral, it is natural to have

$$\oint_w dz a(z)b(w) = \oint_{C_1} dz a(z)b(w) - \oint_{C_2} dz b(w)a(z) = [A, b(w)], \quad (5.19)$$

where

$$A(z) \equiv \oint a(z) dz. \quad (5.20)$$

Here we forced the difference of the radii of circles to approach zero. More generally, we have

$$[A, B] = \oint dw \oint_w dz a(z)b(w), \quad (5.21)$$

where

$$A = \oint a(z)dz, \quad B = \oint b(z)dz. \quad (5.22)$$

The equation (5.21) can be viewed as the integral form of operator product expansion.

State-Operator Correspondence

In a typical quantum field theory, the states and local operators are very different objects. While local operators live at a point in spacetime, the states live over an entire spatial slice. In conformal field theory, states and fields are equivalent, there exists an isomorphism between them. Here we explain the notion of state-operator correspondence.

In the procedure of radial quantization, since the time evolution corresponds to dilation, the generator $L_0 + \bar{L}_0$ is Hamiltonian. To present explicitly, we write the radial quantization in polar coordinate as $z = re^{i\theta}$. Then it is easy to derive following relations:

$$\begin{aligned} L_n &= -\frac{1}{2}r\partial_r + \frac{1}{2}\partial_\theta, \\ \bar{L}_n &= -\frac{1}{2}r\partial_r - \frac{1}{2}\partial_\theta. \end{aligned} \quad (5.23)$$

Furthermore, we have

$$\begin{aligned} L_0 + \bar{L}_0 &= -r\partial_r, \quad \text{dilation} \\ L_0 - \bar{L}_0 &= -i\partial_\theta, \quad \text{rotation.} \end{aligned} \quad (5.24)$$

It turns out the eigenvalues of L_0 are energy, and thus it should be bounded from below. Indeed, unitarity guarantees the lower bound of the eigenvalues of $L_0 + \bar{L}_0$. Additionally, the eigenvalues of $L_0 - \bar{L}_0$ are interpreted as spin.

The Hilbert Space

The Hilbert space of conformal field theory has intricate structure that we will discuss in next chapter. For the moment let's give some facts.

The vacuum must be invariant under conformal transformations, which means it must be annihilated by L_{-1} , L_0 and L_1 :

$$\begin{aligned} L_n |0\rangle &= 0 \\ \bar{L}_n |0\rangle &= 0. \end{aligned} \quad (5.25)$$

We conclude that

$$L_0 |h, \bar{h}\rangle = h |h, \bar{h}\rangle, \quad \bar{L}_0 |h, \bar{h}\rangle = \bar{h} |h, \bar{h}\rangle. \quad (5.26)$$

$|h, \bar{h}\rangle$ is an eigenstate of the Hamiltonian. Similarly, we have

$$L_n |h, \bar{h}\rangle = 0, \quad \bar{L}_0 |h, \bar{h}\rangle = 0 \quad (n > 0). \quad (5.27)$$

Consider a special case of Virasoro algebra:

$$[L_0, L_m] = mL_m, \quad (5.28)$$

this means that L_{-m} increases the conformal dimension. Therefore, excited states can be obtained by successive application of these operators on state $|h\rangle$:

$$L_{-k_1} L_{-k_2} \cdots L_{-k_n} |h\rangle, \quad (1 \leq k_1 \leq k_2 \leq \cdots k_n) \quad (5.29)$$

This state is the eigenstate of L_0 with the eigenvalue

$$h' = h + k_1 + k_2 + \cdots + k_n \equiv h + N. \quad (5.30)$$

The states (5.29) are descendants of the asymptotic state $|h\rangle$ and N is the **level** of the descendant. The number of linearly independent states at level N is the partition number $p(N)$ that can be derived from the generating function:

$$\frac{1}{\varphi(q)} \equiv \prod_{n=1}^{\infty} \frac{1}{1-q^n} = \sum_{n=0}^{\infty} p(n)q^n, \quad (5.31)$$

where $\varphi(q)$ is the Euler function.

6 Minimal Models

6.1 Verma Modules

In quantum mechanics, we are familiar with theories of angular momentum. Recall that the operator J_+ annihilates the state $|j\rangle$. Here we call the state $|j\rangle$ the **highest-weight state**. The other eigenstates of J_z are obtained by applying J_- repeatedly on $|j\rangle$.

In a similar manner, we denote by $|h\rangle$ the highest-weight state, with eigenvalue h of L_0 :

$$L_0 |h\rangle = h |h\rangle, \quad (6.1)$$

where h is the conformal dimension. Operators L_n with positive n annihilates $|h\rangle$:

$$L_n |h\rangle = 0, \quad (n > 0). \quad (6.2)$$

The basis for the other states of the representation, the so-called **descendant states**, is obtained by applying the raising operators in all possible ways:

$$L_{-k_1} L_{-k_2} \cdots L_{-k_n} |h\rangle, \quad (1 \leq k_1 \leq \cdots \leq k_n). \quad (6.3)$$

The space spanned by these bases is called **Verma module**. On the Verma module we define an inner product according to $L_m^\dagger = L_{-m}$. Thus, the inner product of two states is simply

$$\langle h | L_{k_m} \cdots L_{k_1} L_{-l_1} \cdots L_{-l_n} | h \rangle. \quad (6.4)$$

This can be evaluated by using Virasoro algebra (5.17). Notice that the inner product of two states vanishes unless they belong to the same level. In other words, two eigenstates of L_0 are orthogonal. Hermiticity also forces h to be real.

Generally, any state χ —other than the highest-weight state—that is annihilated by $L_n (n > 0)$ is called a **singular vector** (or **null vector**). Such a state generates its own Verma module included in the original. Thus, by this we mean that there is a submodule that it is itself a full-fledged representation of Virasoro algebra. By quotienting out of $V(c, h)$ the null submodule, we can construct **irreducible** representation of Virasoro algebra. Such representations are building blocks of minimal models. Singular vectors are orthogonal to the whole Verma module:

$$\langle \chi | L_{-k_1} \cdots L_{-k_n} | h \rangle = 0, \quad (6.5)$$

so do their descendants.

Virasoro Characters

We associate a generating function $\chi_{c,h}(\tau)$ to a Verma module $V(c, h)$, called the **character** of the module, defined as

$$\chi_{(c,h)}(\tau) = \text{Tr } q^{L_0 - c/24} = \sum_{n=0}^{\infty} (n + h) q^{n+h-c/24}, \quad (6.6)$$

where $q \equiv e^{2\pi i \tau}$. The significance of introducing the factor $q^{-c/24}$ will be evident later. Characters $\bar{\chi}_{(c,\bar{h})}(\bar{\tau})$ can be defined in the same manner. Note that this function is a special function determined solely by the Virasoro algebra, independent of the details of the theory.

The generating function for the partition number at level n is

$$\frac{1}{\varphi(q)} \equiv \prod_{n=1}^{\infty} \frac{1}{1 - q^n} = \sum_{n=0}^{\infty} p(n) q^n, \quad (6.7)$$

the Virasoro character can be written as

$$\chi_{(c,h)}(\tau) = \frac{q^{h-c/24}}{\varphi(q)}. \quad (6.8)$$

In terms of Dedekind function

$$\eta(\tau) \equiv q^{1/24} \varphi(q) = \prod_{n=1}^{\infty} (1 - q^n), \quad (6.9)$$

the Virasoro character becomes

$$\chi_{(c,h)}(\tau) = \frac{q^{h+(1-c)/24}}{\eta(\tau)}. \quad (6.10)$$

The character of irreducible representations of Virasoro algebra is not simply given by this.

6.2 Unitarity and Kac Determinant

We have completed the mathematical formulation of our theory. However, to fully understand its structure, we must impose physical constraints and we shall see the theory's properties strongly depend on the central charge c of Virasoro algebra.

A representation of the Virasoro algebra is considered **unitary** if it contains no negative-norm states, aligning precisely with the probabilistic interpretation of quantum mechanics. In other words, the inner products of states within the Verma module must *not* be negative in any possible configuration. For instance, a unitary bound on (h, c) is obtained by the norm of the state $L_{-n} |h\rangle$:

$$\begin{aligned} \langle h | L_n L_{-n} | h \rangle &= \langle h | \left(L_{-n} L_n + 2nL_0 + \frac{1}{12}cn(n^2 - 1) \right) | h \rangle \\ &= \left[2nh + \frac{1}{12}cn(n^2 - 1) \right] \langle h | h \rangle. \end{aligned} \quad (6.11)$$

Roughly speaking, if $c < 0$ the above becomes negative for sufficiently large n . Therefore representations with negative central charge are illegal.

To analyze this condition more in depth, we proceed as follows: we take all states at a given level, arrange them into a column vector, and construct their conjugate transposes as a row vector. This yields a matrix, formed by their cross products, whose elements correspond to the inner products of the given states. The matrix is referred to as **Kac matrix**. For instance, at level 2, we have 2 basis $L_{-2} |h\rangle$ and $L_{-1}^2 |h\rangle$, then the Kac matrix is

$$M^{(2)} = \begin{pmatrix} \langle h | L_2 L_{-2} | h \rangle & \langle h | L_1^2 L_{-2} | h \rangle \\ \langle h | L_2 L_{-1}^2 | h \rangle & \langle h | L_1^2 L_{-1}^2 | h \rangle \end{pmatrix} = \begin{pmatrix} 4h(2h+1) & 6h \\ 6h & 4h + c/2 \end{pmatrix}. \quad (6.12)$$

The product of the two eigenvalues of $M^{(2)}$ is equal to its determinant:

$$\begin{aligned} \det M^{(2)} &= 32h^3 - 20h^2 + 4h^2c + 2hc \\ &= 32(h - h_{1,1})(h - h_{1,2})(h - h_{2,1}) \end{aligned} \quad (6.13)$$

wherein

$$\begin{aligned} h_{1,1} &= 0 \\ h_{1,2} &= \frac{1}{16} (5 - c - \sqrt{(1-c)(25-c)}) \\ h_{2,1} &= \frac{1}{16} (5 - c + \sqrt{(1-c)(25-c)}). \end{aligned} \quad (6.14)$$

What we have done is just factoring out roots of the determinant. The sum of the eigenvalues is equal to the trace:

$$\text{Tr } M^{(2)} = 8h(h+1) + \frac{2}{c}. \quad (6.15)$$

The representation is not unitary whenever $\det M^{(2)}$ or $\text{Tr } M^{(2)}$ is negative. We have already known that every point (h, c) lying outside the first quadrant leads to a non-unitary representation, then what's the case for points inside? As shown in figure 3, the roots $h_{1,2}$ and $h_{2,1}$ describe two curves that join at $c = 1$. In region surrounded by these two curves, Kac determinant (6.13) is negative and thus the representation is not unitary. Other vanishing curves are drawn as well.

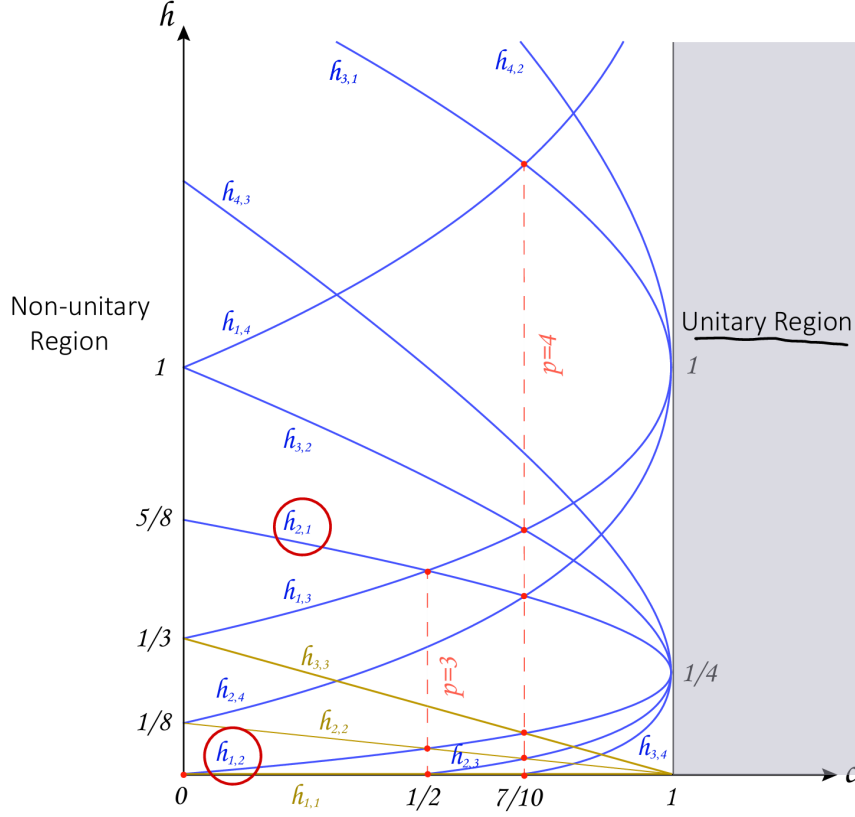


Figure 3: The vanishing curves $C_{r,s}$ for level 2, 3 and 4. The values of r, s are indicated near the curves when they first appear. Source: [6]

There is a general formula for Kac determinant:

$$\det M^{(l)} = \alpha_l \prod_{r,s \geq 1} [h - h_{r,s}(c)]^{P(l-rs)} \quad (6.16)$$

where $P(l - rs)$ is the number of partitions of the integer $l - rs$ and α_l is a positive constant:

$$\alpha_l = \prod_{r,s \geq 1} [(2r)^s s!]^{m(r,s)} \quad (6.17)$$

$$m(r,s) = P(l - rs) - P(l - r(s+1)).$$

The vanishing points of Kac determinant is given by:

$$h_{r,s}(p, p') = \frac{(rp' - sp)^2 - (p - p')^2}{4pp'} \quad (6.18)$$

$$c = 1 - \frac{6(p - p')^2}{pp'} \quad (6.19)$$

The functions $h_{r,s}(c)$ may be expressed as:

$$\begin{aligned} h_{r,s} &= h_0 + \frac{1}{4}(r\alpha_+ + s\alpha_-)^2 \\ h_0 &= \frac{1}{24}(c-1) \\ \alpha_{\pm} &= \frac{\sqrt{1-c} \pm \sqrt{25-c}}{\sqrt{24}}. \end{aligned} \tag{6.20}$$

You may have noticed that $c > 1$ leads to no real roots for $h_{r,s'}$ as shown in figure 3. Hence all representations are unitary. That's why we just focus on the case for $0 < c < 1$. As you can see, the unitarity imposes strong constraints on c . Indeed, a specific value of c corresponds to a CFT.

As stated, the points in the region surrounded by $h_{1,2}$ and $h_{2,1}$ should all be discarded. This is not the end of our discussion because cases of all levels should also be considered. Evidently the Kac determinant changes sign when crossing each vanishing curve. Therefore, a relatively easier way to analyze is to choose a point A in the region with $0 < c < 1$, then connect it with a point B in $c > 1$. If the line AB crosses the vanishing curve odd times, the A is non-unitary; otherwise, it is unitary. Repeatedly performing this procedure to higher levels, you may find that nearly the whole region is discarded, except for vanishing curves, they survived.

Rational Conformal Field Theory

Up to now, we have just applied a weak constraint that Kac determinant must be positive to ensure the unitarity. However, not all theories on the vanishing curves are what we want, equations (6.18) and (6.19) are necessary but not sufficient conditions for unitarity. Let's discuss something simple: theories that lie on the intersection of two curves.

This means that vanishing points of different levels coincide:

$$h_{r,s} = h_{r',s'}. \tag{6.21}$$

With equation (6.18), we have

$$(r-r')p' = (s-s')p \quad \text{or} \quad (r+r')p' = (s+s')p. \tag{6.22}$$

From above, it is obvious to conclude the p and p' must be coprime, i.e. the ratio p/p' is rational. This gives

$$h_{r,s} = h_{r+p,s+p'} \quad \text{and} \quad h_{r,s} = h_{p-r,p'-s}. \tag{6.23}$$

The above equation leads to periodicity and symmetry in CFT.

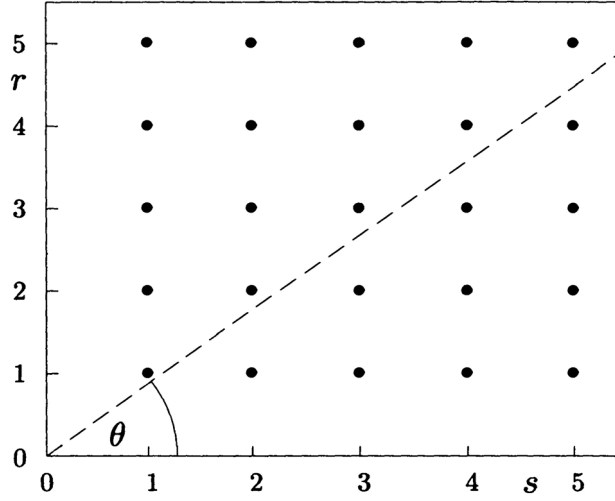


Figure 4: The slope of the line is a rational number.

As shown in 4, the squared-distance between a point (r, s) and a line with slope $\tan \theta$ is

$$\delta = \frac{(p'r - ps)^2}{p'^2 + p^2}, \quad (6.24)$$

therefore we can express the conformal dimension as

$$h_{r,s} = \frac{\delta^2(p'^2 + p^2) - (p - p')^2}{4pp'}, \quad (6.25)$$

namely h depends only on the distance δ . Here we are discussing the **rational conformal field theory**. To constrain the operators in a block, the closeness of operator product expansion is required. Later we will focus on this, the so-called fusion rules.

6.3 Examples

The Yang-Lee Singularity

Ising Model

Potts Model

7 Vertex Operators

7.1 Vertex Operator Algebra

8 Renormalization Group

8.1 Idea of Renormalization Group

As physicists, we study theories of atoms. While for chemists, they care molecules. Theories on different sizes differs a lot. This brings in the idea of renormalization. We now have our eyes back on the critical phenomena. As we have stated, the physics near the critical point can be described by conformal field theory. To obtain a better and more comprehensive understanding, it's crucial to introduce the notion of renormalization, for the reason that the correlation length diverges, making it essential study long-wavelength physics. The central idea of renormalization is to see how physics is affected by integrating out the short-wavelength degrees of freedom. We shall start with a continuous model, and illustrate via path integral formalism.

Consider the following path integral of scalar field ϕ :

$$Z = \int \mathcal{D}\phi e^{iS[\phi]}. \quad (8.1)$$

We introduce a momentum cutoff Λ , in other words, only momentum modes $|k| < \Lambda$ are preserved while others are integrated out. The expression is now denoted as

$$Z = \int \mathcal{D}_\Lambda \phi e^{iS_\Lambda[\phi]}. \quad (8.2)$$

By modifications of Λ , some physical observables, such as correlation functions, will gradually change. However, the core principle here is to keep the action S invariant, which gives

$$\int \mathcal{D}_\Lambda \phi e^{iS_\Lambda[\phi]} = \int \mathcal{D}_{\bar{\Lambda}} \phi e^{iS_{\bar{\Lambda}}[\phi]}, \quad (8.3)$$

then the theory is said to be renormalizable. The transformation from $S_\Lambda[\phi]$ to $S_{\bar{\Lambda}}[\phi]$ is called **renormalization group transformation**, though indeed these transformations do not form a group, due to the lack of inverse.

The process of renormalization can be thought of as a process of coarse-graining: imagine a theory defined by a lattice model with spins distributed on sites, when spins of neighboring sites are averaged out, we become knowing less about the theory. Consequently, the coupling constant between sites must be modified accordingly. To summarize, a renormalization-group transformation amounts to a scale transformation applied both to the integral measure and action.

Renormalization Group Equation

We write the action in a more explicit form:

$$S = - \int d^d x g^i O_i(x), \quad (8.4)$$

where $O_i(x)$ are operators depending on spacetime coordinates and g_i are corresponding coupling constants. For the action to be invariant, coupling constants must change, in order to compensate the change in momentum cutoff Λ . The equation governing the evolution of coupling constants is referred to as the **renormalization group equation**, and it is often written as

$$\frac{\partial g_i}{\partial \ln \Lambda} = \beta^i(g^i). \quad (8.5)$$

The right-hand side is specifically called **beta functions**. Think of Λ as a dynamical variable, the beta functions determine the flow of renormalization group. A well-known example comes from

4-dimensional quantum chromodynamics (QCD). The Lagrangian is given by

$$\mathcal{L} = \frac{1}{4g_{\text{YM}}} \text{Tr}(F^{\mu\nu} F_{\mu\nu}) - i \sum_i \bar{\psi} \not{D} \psi, \quad (8.6)$$

where g_{YM} is the Yang-Mills coupling constant. The beta function to the first order is

$$\beta(g_{\text{YM}}) = -\frac{9}{16\pi^2} g_{\text{YM}}^3. \quad (8.7)$$

This negative beta function makes QCD a theory that is distinctive from others, endowing it with a unique property called **asymptotic freedom**[7], which brilliantly describes phenomenon of quark confinement. This great breakthrough brought Gross, Politzer and Wilczek the Nobel Prize in Physics 2004.

8.2 Fixed Points and Critical Phenomena

For non-trivial coupling constants at critical point, we have

$$\beta^a(g^b) = 0. \quad (8.8)$$

Solutions to this equation $g^b = g_*^b$ are **fixed points** of renormalization group. Linearizing the renormalization group equation around the fixed point, we obtain:

$$\beta^a(g^b) = -\gamma_b^a (g^b - g_*^b). \quad (8.9)$$

Now, consider the eigenvectors $\tilde{g}^a$ associated with the eigenvalues y_a of γ_b^a . Using these eigenvectors as a basis, we can rewrite the action as $\int d^d x \tilde{g}^a \tilde{O}_a(x)$. Since the action itself must be dimensionless under scale transformations, and the coordinates appearing in the d -dimensional integral scale as the inverse of the cutoff, the local operator $\tilde{O}_a$ acquires a scaling dimension:

$$\Delta_a = d - y_a. \quad (8.10)$$

The scaling dimension quantifies how the field behaves under rescaling. Operators are classified according to the sign of y_a :

- If $y_a > 0$ ($\Delta_a < d$), the corresponding local operator is called a **relevant operator**.
- If $y_a < 0$ ($\Delta_a > d$), the operator is an **irrelevant operator**.
- If $y_a = 0$ ($\Delta_a = d$), the operator is **marginal**.

The relevant operator grows under the renormalization-group flow; on the contrary, the irrelevant operator decreases; whereas the marginal operator does not scale with a power law behavior near the critical point, but rather logarithmically. At low-energy regime, it is relevant operators that dominates the theory. This explains why there are a finite number of operators to be fine-tuned to observe the critical phenomena, while irrelevant operators decays rapidly near the critical point.

Consider a scalar field $\phi(x)$ in a spacetime in d dimensions. The classical mass dimension can be derived from the action

$$S = \int d^d x \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi + \frac{1}{2} m^2 \phi^2 \right). \quad (8.11)$$

For the action to be dimensionless, the mass dimension of field ϕ is given by

$$[\phi] = \frac{d-2}{2}, \quad (8.12)$$

and its scaling dimension (in the free theory) is typically the same as its mass dimension. This forbids us to introduce a term, like ϕ^6 , to the action in high-dimensional theory. As the equation above, the mass dimension of this term would be $[\phi^6] = 3(d-2)$, leading to $[\phi^6] > d$ if $d > 3$. The operators become less relevant at low energy scales and hence this term should be discarded.

8.3 Zamolodchikov c-theorem

In renormalization group in 2-dimensional space, there is an important theorem due to Zamolodchikov, called **c-theorem**[8][9], the name c exactly represents central charge. The idea of the c-theorem is to stand back and look at the space of all theories and the renormalization group flows between them. It asserts that, under the assumptions of rotational invariance, unitarity and energy-momentum conservation, there exists a monotonically decreasing function $C(g_i, \mu)$ under the renormalization group flow. Moreover, at the IR-fixed point specified by a set of couplings g_i^* , it is equal to the central charge c .

Hence, one can think that the central charge c encodes degrees of freedom of the theory, and it decreases under a renormalization group since massive particles are integrated out. The theory also formalizes the notion that theories at high energies have more degrees of freedom than theories at low energies and that information is lost as we flow from the former to the latter.

There is also a theorem in four dimensions[10]. In six dimensions, the corresponding theorem is proven for supersymmetric theories[11]. The c-theorem can also be proved in thermodynamics, let us have a look at it.

Given the energy $E = E(S, V)$, the first law of thermodynamics tells us

$$dE = TdS - PdV. \quad (8.13)$$

The free energy F is defined as $F(T, V) = E - TS$ and satisfies

$$dF = -SdT - PdV. \quad (8.14)$$

The free energy is extensive, which gives the fact that it has to be proportional to the volume, therefore

$$F(T, V) = -P(T)V, \quad (8.15)$$

and this in turn gives

$$dF = -\frac{\partial P}{\partial T}VdT - PdV. \quad (8.16)$$

The speed of sound is given by

$$c_s^2 = \frac{dP}{d\epsilon}, \quad (8.17)$$

where $\epsilon = E/V$ is the energy density. At a fixed volume, according to equations (8.13) and (8.17), we have

$$d\epsilon = Tds, \quad (8.18)$$

here $s = dP/dT$ is the entropy density. All of which means that the speed of sound can be expressed as

$$c_s^2 = \frac{1}{T} \frac{dP}{ds} = \frac{1}{T} \frac{dP}{dT} \frac{dT}{ds} = \frac{s}{T} \frac{dT}{ds} = \frac{d \log T}{d \log S}. \quad (8.19)$$

Now we define the c-function as

$$\chi = \frac{s}{T}. \quad (8.20)$$

If we flow from a CFT in the UV, with a central charge c_{UV} , to a different one in the IR with a central

charge c_{IR} , then χ will interpolate between these to central charges (multiplied by $8\pi^2/3$). To prove the c theorem, we need to show that as we decrease the temperature, and so excite lower energy degrees of freedom, the function χ necessarily decreases. We can do this by relating χ with the speed of sound

$$\frac{1}{c_s^2} = \frac{d \log T}{d \log S} = \frac{d \log(\chi T)}{d \log S} = 1 + \frac{d \log \chi}{d \log T}. \quad (8.21)$$

By causality, we must have $c_s^2 \leq 1$ (with equality when we have a CFT) and so

$$\frac{d \log \chi}{d \log T} \geq 0 \quad \Rightarrow \quad \frac{d \chi}{d T} \geq 0. \quad (8.22)$$

This is exactly what we wanted. We learn that we necessarily have $c_{UV} \geq c_{IR}$. This is the c-theorem. The proof above is not complete to some extent. Refer to lecture notes written by David Tong.

9 Modular Invariance

9.1 Conformal Field Theory on the Torus

The modular invariance on the torus imposes constraints on the operator of the theory. A torus is defined by specifying two linearly independent lattice vectors on a plane and identifying points that differ by integer combinations of these vectors. Before the construction of the theory, we should be well-prepared for mathematical foundations. Now we consider the lattice generated by vectors ω_1 and ω_2 . The conformal field theory defined on a torus do not depend on the overall scale of the lattice, nor on the absolute orientation of the lattice vectors. The relevant parameter is **modular parameter**, defined as $\tau = \omega_1/\omega_2$. Evidently τ is complex. Translating a point as

$$z \rightarrow z + m\omega_1 + n\omega_2, \quad m, n \in \mathbb{Z} \quad (9.1)$$

and identifying the two points before and after translation as the equivalence, we obtain a torus.

The choice of basis is not unique: $(\omega_1, \omega_1 + \omega_2)$ and $(\omega_1 + \omega_2, \omega_1)$ generate the same torus. The complex parameter transforms along as

$$\begin{aligned} \mathcal{T} : \tau &\rightarrow \tau + 1 \\ \mathcal{U} : \tau &\rightarrow \frac{\tau}{\tau + 1}. \end{aligned} \quad (9.2)$$

S transformation is defined as

$$\mathcal{S} : \tau \rightarrow -\frac{1}{\tau}. \quad (9.3)$$

Above transformations give $\mathcal{U} = \mathcal{T}\mathcal{S}\mathcal{T}$. For a general transformation

$$\begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} \rightarrow \begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}, \quad (9.4)$$

to ensure the basis to generate the same lattice, a, b, c and d must all be integers and $ad - bc = 1$. In other words, the linear transformation matrix must be in the **modular group** $PSL(2, \mathbb{Z}) = SL(2, \mathbb{Z})/\mathbb{Z}_2$. We also refer to the corresponding transformation as **modular transformation**. The modular parameter transforms as

$$\tau \rightarrow \frac{a\tau + b}{c\tau + d}. \quad (9.5)$$

The group $PSL(2, \mathbb{Z})$ is generated by transformation T and S , meanwhile it is with ease to verify that matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (9.6)$$

satisfy $S^2 = (ST)^3 = 1$. By combining the $\mathcal{S}$ and $\mathcal{T}$ transformations, we can derive various conformal bootstrap equations from the partition function. These are specifically called **modular bootstrap** equations.

However, you may have been wondering why do we need a theory on the torus? The essential distinction from the infinite plane is the occurrence of periodic conditions. This does not necessarily mean that the fields themselves are simply periodic. For instance, a real fermion living on a torus may pick a -1 factor when translated by a period: this constitutes Neveu-Schwarz (NS) condition, whereas the Ramond (R) condition demands that the fermion is periodic. There are in all four types of boundary conditions combined by these two.

With mathematical preparations, we shall treat the theory in the operator formalism, the relevant quantity is partition function Z and its dependence on the modular parameter τ . We let H and P

denote the Hamiltonian and momentum of the theory, respectively, then the operator that translates the system parallel to the period ω_2 over a distance a in Euclidean spacetime is

$$\exp\left(-\frac{a}{|\omega_2|}(H\text{Im}\omega_2 - iP\text{Re}\omega_2)\right)$$

The above translation takes us to from one row of the lattice to the next, parallel to the period ω_2 . The partition function is obtained by taking the trace to the translation operator:

$$Z(\omega_1, \omega_2) = \text{Tr}[\exp(-H\text{Im}\omega_2 - iP\text{Re}\omega_2)]. \quad (9.7)$$

We can express the operators H and P in terms of Virasoro characters L_0 and $\bar{L}_0$:

$$\begin{aligned} H &= \frac{2\pi}{L} \left(L_0 + \bar{L}_0 - \frac{c}{12} \right) \\ P &= \frac{2\pi i}{L} (L_0 - \bar{L}_0). \end{aligned} \quad (9.8)$$

The final partition function can be written as

$$Z(\tau) = \text{Tr} \left(q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - c/24} \right), \quad (9.9)$$

where

$$q = \exp 2\pi i \tau, \quad \bar{q} = \exp(-2\pi i \bar{\tau}). \quad (9.10)$$

Note that partition function depends on the periods $\omega_{1,2}$ only by the ratio τ . We expect that the partition function will be expressible as a bilinear combination of characters of the Verma modules forming the Hilbert space of the theory.

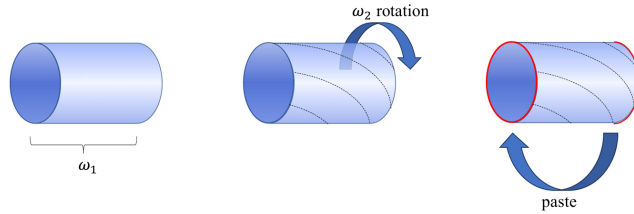


Figure 5: Construction of partition funtion. Figure from:[12]

It is worth notifying that under $\mathcal{T}$ transformation, the partition changes as

$$Z(\tau + 1) = \text{Tr} \left(q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - c/24} \right) e^{2\pi i(L_0 - \bar{L}_0)}. \quad (9.11)$$

Since the eigenvalues of $L_0 - \bar{L}_0$ correspond to spin, the invariance under modular $\mathcal{T}$ transformation implies that the spin must be integer.

9.2 Free Bosons and Fermions on the Torus

9.3 Orbifold Partition Function

As we have seen in previous examples, the theory strongly depends on the manifold it resides. In particular, the boundary condition plays an significant role. What we want to consider now is called **orbifold**. Roughly speaking, orbifold is the topological space whose quotient space of finite groups

are locally Euclidean. The simplest example of orbifold is constructed by identifying points x and $-x$ on a circle, which leads to a finite line. The resulting orbifold is denoted as $S^1/\mathbb{Z}_2$.

Then, the modular invariant partition function of a free boson on the $\mathbb{Z}_2$ -orbifold reads

$$Z_{\text{orb}}(\tau, \bar{\tau}) = \frac{1}{2} Z_{\text{R}}(\tau, \bar{\tau}) + \left| \frac{\eta(\tau)}{\vartheta_2(\tau)} \right| + \left| \frac{\eta(\tau)}{\vartheta_3(\tau)} \right| + \left| \frac{\eta(\tau)}{\vartheta_4(\tau)} \right|. \quad (9.12)$$

10 WZW Models

10.1 Nonlinear Sigma models

10.2 Affine Lie Algebra

Properly defining the WZW model requires an explanation of the affine Lie algebra. Though it may seem to deviate from the central contents of notes, we try to give a relatively simple introduction here.

10.3 Wess-Zumino-Witten Model

Generally we consider a CFT with level- k affine Lie algebra $\hat{\mathfrak{g}}$ symmetry,

$$[J_n^a, J_m^b] = \sum_c i f_{abc} J_{n+m}^c + kn \delta_{ab} \delta_{n+m,0}, \quad (10.1)$$

where J_n^a are the n -th mode of the current, defined as

$$j_n^a = \oint \frac{dz}{2\pi i} z^n j_a(z). \quad (10.2)$$

This is indeed the **Kac-Moody algebra**. Note that the zeroth mode reduces the equation to the ordinary Lie algebra. By generalizing $\hat{\mathfrak{u}}_1$ to non-Abelian, new terms with non-trivial structure constants f_{abc} appear.

10.4 Sugawara Construction

10.5 Fusion Rules

11 AdS/CFT Correspondence

The AdS/CFT correspondence [13] is a special case of holographic principle, which sets up the equivalence between a $(d + 1)$ -dimensional gravity theory and a d -dimensional CFT. The correspondence has been widely generalized though it was firstly proposed to describe string theory.

11.1 Anti-de Sitter space

Anti-de Sitter space is a maximally symmetric manifold with constant negative scalar curvature.

11.2 Entanglement Entropy

11.3 GKPW Dictionary

A Lie Group and Lie Algebra

A **Lie algebra** $\mathfrak{g}$ is a vector space whose elements can be combined by a second law, the Lie bracket, which we represent by $[A, B] = C$ with A, B, C elements of $\mathfrak{g}$, in such a way that the law is bilinear, skew-symmetric and satisfies the well-known Jacobi identity [blanes2009magnus]

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0. \quad (\text{A.1})$$

When dealing with matrices we take, as Lie bracket, the familiar commutator:

$$[A, B] = AB - BA, \quad A, B \in \mathfrak{g}. \quad (\text{A.2})$$

If we consider a finite-dimensional Lie algebra with dimension d and denote by $A_i, i = 1, \dots, d$, the vectors of one of its bases, then the fundamental brackets one has to know are

$$[A_i, A_j] = c_{ijk} A_k, \quad (\text{A.3})$$

where Einstein's summation convention is understood. The coefficients c_{ijk} are so-called structure constants of the algebra.

Associated with any $A \in \mathfrak{g}$ we can define a linear operator $\text{ad}_A : \mathfrak{g} \rightarrow \mathfrak{g}$ which acts according to

$$\text{ad}_A B = [A, B], \quad \text{ad}_A^j B = [A, \text{ad}_A^{j-1} B], \quad \text{ad}_A^0 B = B, \quad j \in \mathbb{N}, B \in \mathfrak{g}. \quad (\text{A.4})$$

In mathematics, the **Baker–Campbell–Hausdorff** formula gives the value of Z that solves the equation

$$e^X e^Y = e^Z \quad (\text{A.5})$$

For possibly *non-commutative* X and Y in the Lie algebra in a Lie group. There are various ways of writing the formula, but all ultimately yield an expression for Z in Lie algebraic terms, that is, as a formal series (not necessarily convergent) in X and Y and iterated commutators thereof. The first few terms of this series are

$$Z = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \dots \quad (\text{A.6})$$

If X and Y are sufficiently small elements of the Lie algebra $\mathfrak{g}$ of a Lie group G , the series is convergent. Another version of BCH formula is

$$e^X Y e^{-X} = Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \dots. \quad (\text{A.7})$$

$e^X e^Y e^{-X} = \exp(e^X Y e^{-X})$ is also an important identity.

B Elliptic and Theta Function Identities

C Liouville Conformal Field Theory

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